

Regression Assignment

1. Machine Learning

Supervised Learning

Regression

2. Total no. of Rows: 1338

Total no. of Columns: 6

3. Nominal Method

One Hot Encoding

4.R2 Values

1.Multiple linear Regression R2 value : 0.789

2.Support Vector Regression R2 Value:

s.no	HYPER PARAMETER	CRITERION (LINEAR)R2	CRITERION (rbf)R2	CRITERION (poly)R2	CRITERION (sigmoid)R2
1	C=100	0.6288	0.3200	0.6179	0.5276
2	C=1000	0.7649	0.8102	0.8566	0.2874
3	C=2000	0.7440	0.8547	0.8605	-0.5979
4	C=3000	0.7414	0.8663	0.8598	-2.1244

Support Vector Regression R2 Value: C=3000,
Criterion(rbf)=0.8663

3.DECISION TREE R2 VALUE:

Decision Tree Regression R2 Value:0.7511

(Poisson,sqrt,Best)

s.no	CRITERION	MAX_FEATURES	SPLITTER	R2VALUE
1	Absolute_error	sqrt	Best	0.6027
2	Absolute_error	sqrt	Random	0.6940
3	Absolute_error	Log2	Best	0.7358
4	Absolute_error	Log2	Random	0.6507
5	squared_error	sqrt	Best	0.7224
6	squared_error	sqrt	Random	0.7092
7	squared_error	Log2	Best	0.7179
8	squared_error	Log2	Random	0.6906
9	Poisson	sqrt	Best	0.7511
10	Poisson	sqrt	Random	0.6437
11	Poisson	Log2	Best	0.7225
12	poisson	Log2	Random	0.6639

4.RANDOM FOREST R2 VALUE:

Random Forest Regression R2 Value=**0.8710**

(squared_error,n_estimators=100,sqrt,log2)

S.NO	CRITERION	n_estimators	max_features	R2 Value
1	squared_error	50	Sqrt,log2	0.8697
2	squared_error	100	Sqrt,log2	0.8710
3	absolute_error	50	Sqrt,log2	0.8657
4	absolute_error	100	Sqrt,log2	0.8692
5	poisson	50	Sqrt,log2	0.8630
6	poisson	100	Sqrt,log2	0.8678

5.The Best Model is

Support Vector Regression R2 Value: C=3000,

Criterion(rbf)=**0.8663**

Random Forest Regression R2 Value=**0.8710**

(squared_error,n_estimators=100,sqrt,log2)

In Random Forest Regression R2 value is 0.8710 ,Which is nearest to 1.So ,This model perform well